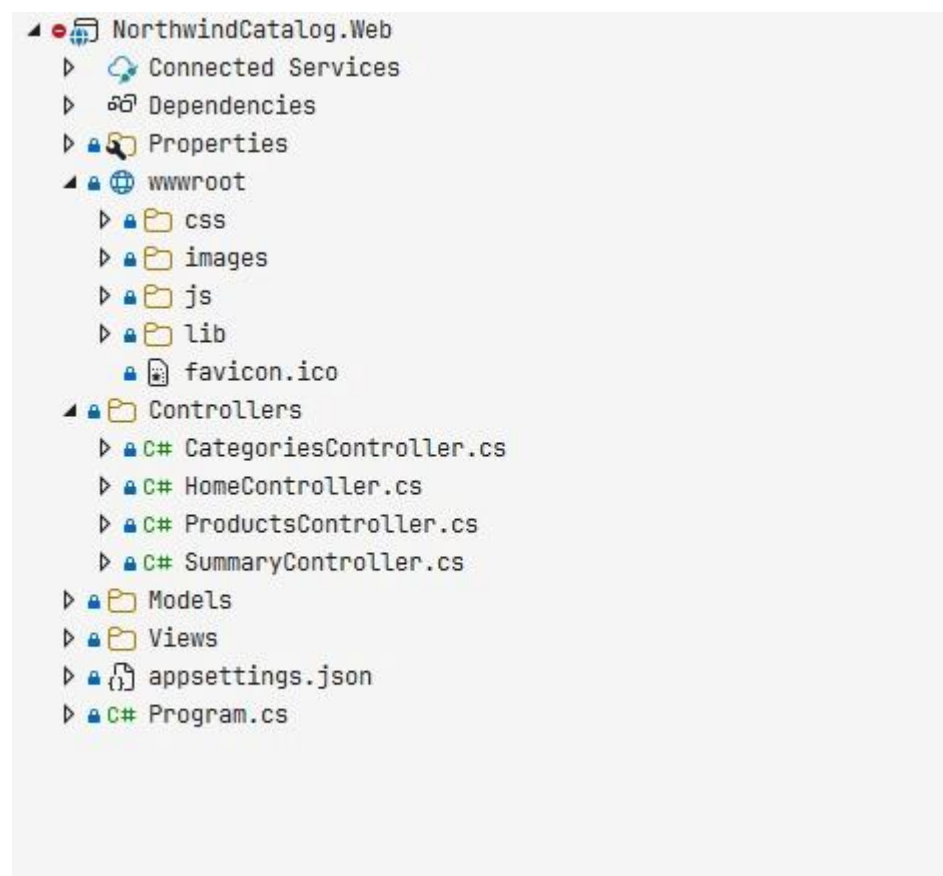


Northwind Catalog – Week 15 Assessment

Name: Krish

The solution consists of three projects:

- **NorthwindCatalog.Services** → **Web API + Business Logic + EF Core**
- **NorthwindCatalog.Web** → **MVC UI (API Consumption)**
- **NorthwindCatalog.Tests** → **Unit Testing using xUnit**



Fluent Api for scaffolding:

Use following commands in Package Manager:

Scaffold-DbContext "Connection_String"

Microsoft.EntityFrameworkCore.SqlServer -OutputDir Models -Tables Categories,Products

The connection string will be present in the appsettings.json file.

```
0 references
protected override void OnConfiguring(DbContextOptionsBuilder optionsBuilder)
⇒ optionsBuilder.UseSqlServer("Server=localhost\\SQLEXPRESS02;Database=Northwind;Trusted_Connection=True;TrustServerCertificate=True;");

0 references
protected override void OnModelCreating(ModelBuilder modelBuilder)
{
    modelBuilder.Entity<Category>(entity ⇒
    {
        entity.HasIndex(e ⇒ e.CategoryName, "CategoryName");

        entity.Property(e ⇒ e.CategoryId).HasColumnName("CategoryID");
        entity.Property(e ⇒ e.CategoryName).HasMaxLength(15);
        entity.Property(e ⇒ e.Description).HasColumnType("ntext");
        entity.Property(e ⇒ e.Picture).HasColumnType("image");
    });

    modelBuilder.Entity<Product>(entity ⇒
    {
        entity.HasIndex(e ⇒ e.CategoryId, "CategoriesProducts");

        entity.HasIndex(e ⇒ e.CategoryId, "CategoryID");

        entity.HasIndex(e ⇒ e.ProductName, "ProductName");

        entity.HasIndex(e ⇒ e.SupplierId, "SupplierID");

        entity.HasIndex(e ⇒ e.SupplierId, "SuppliersProducts");

        entity.Property(e ⇒ e.ProductId).HasColumnName("ProductID");
        entity.Property(e ⇒ e.CategoryId).HasColumnName("CategoryID");
        entity.Property(e ⇒ e.ProductName).HasMaxLength(40);
        entity.Property(e ⇒ e.QuantityPerUnit).HasMaxLength(20);
        entity.Property(e ⇒ e.ReorderLevel).HasDefaultValue((short)0);
        entity.Property(e ⇒ e.SupplierId).HasColumnName("SupplierID");
        entity.Property(e ⇒ e.UnitPrice)
            .HasDefaultValue(0m)
            .HasColumnType("money");
        entity.Property(e ⇒ e.UnitsInStock).HasDefaultValue((short)0);
        entity.Property(e ⇒ e.UnitsOnOrder).HasDefaultValue((short)0);

        entity.HasOne(d ⇒ d.Category).WithMany(p ⇒ p.Products)
            .HasForeignKey(d ⇒ d.CategoryId)
            .HasConstraintName("FK_Products_Categories");
    });

    OnModelCreatingPartial(modelBuilder);
}
```

Services API Swagger:

The image shows a Visual Studio Solution Explorer on the left and the Swagger UI on the right.

Solution Explorer:

- Solution 'NorthwindCatalog' (3 of 3 projects)
 - GitHub Actions
 - NorthwindCatalog.Services
 - Connected Services
 - Dependencies
 - Properties
 - launchSettings.json
 - Controllers
 - CategoryController.cs
 - ProductController.cs
 - WeatherForecastController.cs
 - Data
 - NorthwindContext.cs
 - DTOs
 - CategoryDto.cs
 - CategorySummaryDto.cs
 - ProductDto.cs
 - Mapping
 - MappingProfile.cs
 - Models
 - Category.cs
 - Product.cs
 - Repository
 - CategoryRepository.cs
 - ICategoryRepository.cs
 - IProductRepository.cs
 - ProductRepository.cs
 - appsettings.json
 - NorthwindCatalog.Services.http
 - Program.cs
 - WeatherForecast.cs
 - NorthwindCatalog.Tests
 - NorthwindCatalog.Web

Swagger UI:

Swagger
Select a definition: NorthwindCatalog.Services v1

NorthwindCatalog.Services 1.0 OAS 3.0
<http://localhost:5000/swagger/v1/swagger.json>

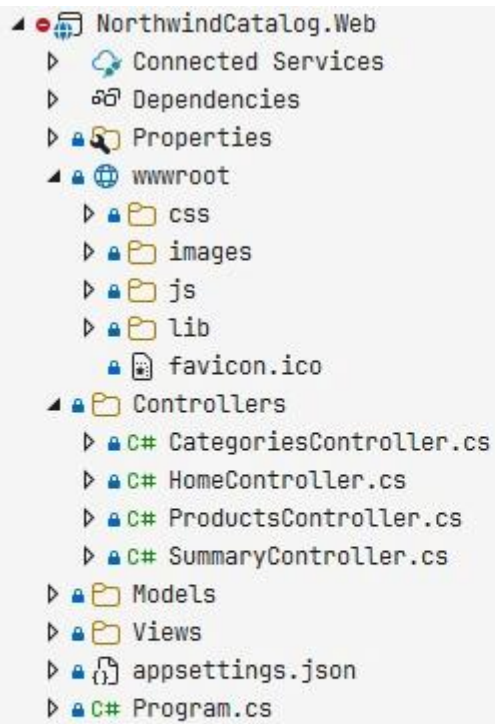
CategoriesApi

- GET /api/categories

ProductsApi

- GET /api/products/by-category/{categoryId}
- GET /api/products/summary

MVC Controllers:



Categories Page:

Categories



Beverages
img : /images/Beverages.jpeg
[View Products](#)



Condiments
img : /images/Condiments.jpeg
[View Products](#)



Confections
img : /images/Confections.jpeg
[View Products](#)



Dairy Products
img : /images/DairyProducts.jpeg
[View Products](#)



Grains/Cereals
img : /images/GrainsCereals.jpeg
[View Products](#)



Meat/Poultry
img : /images/MeatPoultry.jpeg
[View Products](#)



Produce
img : /images/Produce.jpeg
[View Products](#)



Seafood
img : /images/Seafood.jpeg
[View Products](#)

Product Detail Page:

Products

Chai
Price: ₹18.0000
Stock: 39
Inventory: ₹702.0000

Chang
Price: ₹19.0000
Stock: 17
Inventory: ₹323.0000

Guaraná Fantástica
Price: ₹4.5000
Stock: 20
Inventory: ₹90.0000

Sasquatch Ale
Price: ₹14.0000
Stock: 111
Inventory: ₹1554.0000

Steeleye Stout
Price: ₹18.0000
Stock: 20
Inventory: ₹360.0000

Côte de Blaye
Price: ₹263.5000
Stock: 17
Inventory: ₹4479.5000

Chartreuse verte
Price: ₹18.0000
Stock: 69
Inventory: ₹1242.0000

Ipoh Coffee
Price: ₹46.0000
Stock: 17
Inventory: ₹782.0000

Laughing Lumberjack Lager
Price: ₹14.0000
Stock: 52
Inventory: ₹728.0000

Outback Lager
Price: ₹15.0000
Stock: 15
Inventory: ₹225.0000

Rhönbräu Klosterbier
Price: ₹7.7500
Stock: 125
Inventory: ₹968.7500

Lakkalikööri
Price: ₹18.0000
Stock: 57
Inventory: ₹1026.0000

Summary Page:

Summary			
Category	Products	Avg Price	Most Expensive
Beverages	12	₹37.979166	Côte de Blaye
Condiments	12	₹23.062500	Veggie-spread
Confections	13	₹25.160000	Sir Rodney's Marmalade
Dairy Products	10	₹28.730000	Raclette Courdavault
Grains/Cereals	7	₹20.250000	Gnocchi di nonna Alice
Meat/Poultry	6	₹54.006666	Thüringer Rostbratwurst
Produce	5	₹32.370000	Manjimup Dried Apples
Seafood	12	₹20.682500	Camaron Tigers

Testing:

